Chapter 2. Simple Comparative Experiments

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1 Basic Concepts

Definition 2.1

- **Population**: all possible items under specified experimental or observational conditions.
- **Sample**: a group of observations taken from a population.
- **Parameter**: a numerical characteristic of a population (e.g. μ, σ^2).
- **Statistic**: a numerical characteristic computed from the sample (e.g. $\bar{y}, S^2$).
- **Mathematical model**: derived from theoretical considerations; represents exact, error-free relationships among variables.

- **Statistical model**: derived from data subject to observation, experimental and/or measurement errors (e.g. $y = \mu + \varepsilon$).

2 Descriptive Statistics

2.1 Traditional statistics

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

2.2 Robust statistics

- **Sample median**: with order statistics $x_{(1)} \le \cdots \le x_{(n)}$, (i) if n is odd, the median is $x_{(q)}$ with $q = (n+1)/2$; (ii) if n is even, it is $[x_{(q)} + x_{(q+1)}]/2$ with $q = n/2$.
- **Median absolute deviation**:

$$\text{MAD} = \frac{\text{median}(|x_i - \text{median}(x)|)}{0.6745}.$$

  The constant $0.6745 = \Phi^{-1}(0.75)$ makes MAD a consistent estimator of σ under normality: if $X \sim N(\mu, \sigma^2)$ then $\text{median}|X - \mu| = 0.6745\sigma$, because $P(|X - \mu| \le c\sigma) = 0.5 \iff 2\Phi(c) - 1 = 0.5 \iff \Phi(c) = 0.75$.

Example 2.2: Robustness of median/MAD

Data 1: $1, 2, 3, 4, 5$.

$$\bar{x} = \tfrac{15}{5} = 3, \qquad \text{median} = 3,$$
$$s^2 = \tfrac{1}{4}[(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2] = \tfrac{10}{4} = 2.5, \quad s = 1.581,$$
$$|x_i - 3| = 2, 1, 0, 1, 2 \;\Rightarrow\; \text{median} = 1, \quad \text{MAD} = 1/0.6745 = 1.482.$$

Data 2: $1, 2, 3, 4, 1000000$ (one gross outlier).

$$\bar{x} = \tfrac{1000010}{5} = 200002, \qquad \text{median} = 3,$$
$$s = 447212.5 \quad \text{(explodes)},$$
$$|x_i - 3| = 2, 1, 0, 1, 999997 \;\Rightarrow\; \text{median} = 1, \quad \text{MAD} = 1.482 \quad \text{(unchanged)}.$$

The mean and standard deviation are destroyed by a single outlier, while median and MAD are unaffected.

3 Graphical Display of Data

- **Histogram**: displays the empirical distribution of (continuous) data.
- **Boxplot**: shows the median, lower and upper quartiles, whiskers and outliers; good for comparing several samples side by side.
- **Normal Q–Q plot**: checks normality (details in Ch. 3).

4 Simple Comparative Experiments: the Mortar Example

- Compare two conditions (treatments) = two levels of one factor.
- **Example**: tension bond strength of Portland cement mortar for two formulations, *modified* and *unmodified*; 10 observations each.

Modified (y_{1j})	16.85	16.40	17.21	16.35	16.52	17.04	16.96	17.15	16.59	16.57
Unmodified (y_{2j})	17.50	17.63	18.25	18.00	17.86	17.75	18.22	17.90	17.96	18.15

- Modified: mean = 16.764, median = 16.72, $s = 0.3164$, MAD = 0.4151.
- Unmodified: mean = 17.922, median = 17.93, $s = 0.2479$, MAD = 0.2965.
- Question: $H_0 : \mu_{\text{mod}} = \mu_{\text{unmod}}$?

5 Two-Sample t-Test

5.1 Model and assumptions

$$y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, 2,\ j = 1, \ldots, n_i,$$

1. $y_{1j} \sim N(\mu_1, \sigma_1^2)$, $y_{2j} \sim N(\mu_2, \sigma_2^2)$, all independent;
2. common variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

Hypotheses: $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$.

5.2 Derivation of the test statistic

Step 1: distribution of $\bar{y}_1 - \bar{y}_2$. Since $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$ is a linear combination of independent normals,

$$\begin{aligned} \mathrm{E}(\bar{y}_1 - \bar{y}_2) &= \mathrm{E}(\bar{y}_1) - \mathrm{E}(\bar{y}_2) = \mu_1 - \mu_2, \\ \mathrm{Var}(\bar{y}_1 - \bar{y}_2) &= \mathrm{Var}(\bar{y}_1) + \mathrm{Var}(\bar{y}_2) - 2\underbrace{\mathrm{Cov}(\bar{y}_1, \bar{y}_2)}_{=0 \text{ (indep.)}} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}. \end{aligned}$$

Hence $\bar{y}_1 - \bar{y}_2 \sim N\Big(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\Big)$.

Step 2: known variances. If σ_1^2, σ_2^2 were known, under H_0

$$Z_0 = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \overset{\sigma_1 = \sigma_2 = \sigma}{=} \frac{\bar{y}_1 - \bar{y}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0, 1).$$

Step 3: pooled variance. With $\sigma_1^2 = \sigma_2^2 = \sigma^2$ unknown, both S_1^2 and S_2^2 estimate the same σ^2; pool them:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Theorem 2.3: S_p^2 is unbiased and $(n_1+n_2-2)S_p^2/\sigma^2 \sim \chi^2_{n_1+n_2-2}$

Proof. *Unbiasedness.* For one normal sample, $\mathrm{E}(S_i^2) = \sigma^2$:

$$\mathrm{E}\Big[\sum_j (y_{ij}-\bar{y}_i)^2\Big] = \mathrm{E}\Big[\sum_j y_{ij}^2 - n_i\bar{y}_i^2\Big] = \sum_j \mathrm{E}(y_{ij}^2) - n_i\mathrm{E}(\bar{y}_i^2)$$
$$= n_i(\sigma^2+\mu_i^2) - n_i\Big(\frac{\sigma^2}{n_i}+\mu_i^2\Big) = (n_i-1)\sigma^2.$$

Hence

$$\mathrm{E}(S_p^2) = \frac{(n_1-1)\sigma^2+(n_2-1)\sigma^2}{n_1+n_2-2} = \sigma^2.$$

Distribution. For a normal sample, $(n_i-1)S_i^2/\sigma^2 \sim \chi^2_{n_i-1}$ and $S_i^2 \perp\!\!\!\perp \bar{y}_i$. The two samples are independent, and the sum of independent χ^2's is χ^2 with the dfs added:

$$\frac{(n_1+n_2-2)S_p^2}{\sigma^2} = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_1-1} + \chi^2_{n_2-1} = \chi^2_{n_1+n_2-2}.$$

■

Step 4: the t statistic. Recall: if $Z \sim N(0,1)$, $V \sim \chi^2_\nu$, $Z \perp\!\!\!\perp V$, then $Z/\sqrt{V/\nu} \sim t_\nu$. With $Z = Z_0$ and $V = (n_1+n_2-2)S_p^2/\sigma^2$ (independent because $(\bar{y}_1,\bar{y}_2) \perp\!\!\!\perp (S_1^2,S_2^2)$):

$$t_0 = \frac{\dfrac{\bar{y}_1-\bar{y}_2}{\sigma\sqrt{1/n_1+1/n_2}}}{\sqrt{\dfrac{(n_1+n_2-2)S_p^2/\sigma^2}{n_1+n_2-2}}} = \frac{\bar{y}_1-\bar{y}_2}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \overset{H_0}{\sim} t_{n_1+n_2-2}.$$

The unknown σ cancels; this is exactly "replace σ by S_p".

- t_0 is a *distance measure*: how far apart the two averages are, in standard-error units.
- Values of t_0 near 0 are consistent with H_0; values far from 0 support H_1.
- **Decision rule**: reject H_0 if $|t_0| > t_{\alpha/2,\,n_1+n_2-2}$.
- ***P*-value**: the smallest level of significance that would lead to rejection of H_0, $P = P(|t_{n_1+n_2-2}| \geq |t_0|)$.

5.3 Confidence interval

From $\dfrac{(\bar{y}_1-\bar{y}_2)-(\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}} \sim t_{n_1+n_2-2}$,

$$1-\alpha = P\left(-t_{\alpha/2} \leq \frac{(\bar{y}_1-\bar{y}_2)-(\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}} \leq t_{\alpha/2}\right)$$
$$= P\left(\bar{y}_1-\bar{y}_2 - t_{\alpha/2}S_p\sqrt{\tfrac{1}{n_1}+\tfrac{1}{n_2}} \leq \mu_1-\mu_2 \leq \bar{y}_1-\bar{y}_2 + t_{\alpha/2}S_p\sqrt{\tfrac{1}{n_1}+\tfrac{1}{n_2}}\right).$$

Example 2.4: Mortar data, line by line

$n_1 = n_2 = 10$, $\bar{y}_1 = 16.764$, $\bar{y}_2 = 17.922$, $S_1^2 = 0.3164^2 = 0.1001$, $S_2^2 = 0.2479^2 = 0.0615$.

$$S_p^2 = \frac{9(0.1001)+9(0.0615)}{18} = \frac{0.1001+0.0615}{2} = 0.0808, \qquad S_p = 0.2843,$$

$$S_p\sqrt{\tfrac{1}{10}+\tfrac{1}{10}} = 0.2843 \times 0.4472 = 0.1271,$$
$$t_0 = \frac{16.764 - 17.922}{0.1271} = \frac{-1.158}{0.1271} = -9.11.$$

Since $|t_0| = 9.11 > t_{0.025,18} = 2.101$, reject H_0: the mean strengths of the two formulations differ. P-value $= P(|t_{18}| \geq 9.11) = 3.67\times10^{-8}$, so H_0 is rejected at any level $\alpha \geq 3.67\times10^{-8}$. 95% CI: $-1.158 \pm 2.101(0.1271) = -1.158 \pm 0.267 = [-1.425,\ -0.891]$ (matches the R output `t.test(..., var.equal=TRUE)`).

5.4 Checking assumptions

- **Normality**: normal Q–Q plot of each sample (`qqnorm`, `qqline`).
- **Equal variances**: F-test. Under $H_0 : \sigma_1^2 = \sigma_2^2$,

$$F_0 = \frac{S_1^2}{S_2^2} = \frac{[(n_1-1)S_1^2/\sigma^2]/(n_1-1)}{[(n_2-1)S_2^2/\sigma^2]/(n_2-1)} \sim F_{n_1-1,\,n_2-1}.$$

  Mortar: $F_0 = 0.1001/0.0615 = 1.629$, $P = 0.4785 \Rightarrow$ no evidence against equal variances.

6 The Case $\sigma_1^2 \neq \sigma_2^2$ (Welch)

Use

$$t_0 = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}},$$

whose null distribution is well approximated by t_ν with (Satterthwaite)

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}}.$$

Remark 2.5: Where ν comes from

Let $W = S_1^2/n_1 + S_2^2/n_2$. We approximate $\nu W/\mathrm{E}(W) \approx \chi^2_\nu$ and choose ν by matching variances. A χ^2_ν has variance 2ν, so we need $\mathrm{Var}(\nu W/\mathrm{E}W) = 2\nu \iff \nu = 2[\mathrm{E}W]^2/\mathrm{Var}(W)$. Using $\mathrm{Var}(S_i^2) = 2\sigma_i^4/(n_i-1)$ (since $(n_i-1)S_i^2/\sigma_i^2 \sim \chi^2_{n_i-1}$):

$$\mathrm{E}W = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \mathrm{Var}W = \frac{2\sigma_1^4}{n_1^2(n_1-1)} + \frac{2\sigma_2^4}{n_2^2(n_2-1)},$$
$$\nu = \frac{2(\sigma_1^2/n_1 + \sigma_2^2/n_2)^2}{2\frac{(\sigma_1^2/n_1)^2}{n_1-1} + 2\frac{(\sigma_2^2/n_2)^2}{n_2-1}},$$

and plugging in S_i^2 for σ_i^2 gives the formula. Mortar: $\nu = 17.03$ (R: Welch $t = -9.11$, df$= 17.025$).

7 Permutation (Randomization) Test

Derivation

Instead of assuming normality, use the randomization itself as the basis of inference.

- Pool the $N = n + m$ observations. Under $H_0 : \mu_x = \mu_y$ (more precisely, under H_0 the two distributions are identical) the labels are exchangeable.
- Form all $N!$ permutations of the data; for each permutation compute the test statistic $T_1, \ldots, T_{N!}$.
- Under H_0 each of these values is equally likely, so the permutation distribution of T puts mass $1/N!$ on each T_j:

$$p\text{-value} = P(T > t_{\text{obs}} \mid H_0) = \frac{1}{N!} \sum_{j=1}^{N!} I(T_j > t_{\text{obs}}).$$

Example 2.6: A tiny permutation test

$(X_1, X_2, Y_1) = (1, 9, 3)$, i.e. x-sample $\{1, 9\}$ $(n = 2)$, y-sample $\{3\}$ $(m = 1)$. Statistic $T = |\bar{x} - \bar{y}|$.

Permutation (X_1, X_2, Y_1)	$\bar{x}$	$T = \lvert\bar{x} - \bar{y}\rvert$	prob.
$(1, 9, 3)$	5	$\lvert 5 - 3\rvert = 2$	1/6
$(1, 3, 9)$	2	$\lvert 2 - 9\rvert = 7$	1/6
$(9, 1, 3)$	5	2	1/6
$(9, 3, 1)$	6	$\lvert 6 - 1\rvert = 5$	1/6
$(3, 9, 1)$	6	5	1/6
$(3, 1, 9)$	2	7	1/6

Observed $t_{\text{obs}} = 2$. Then $P(T \geq 2) = 6/6 = 1$ and $P(T > 2) = 4/6$: no evidence against H_0 (and with $N = 3$ the smallest attainable p-value is $2/6$, so such a test can never reject at $\alpha = 0.05$).

8 Paired Comparison Design

8.1 Hardness testing example

- A hardness testing machine presses a rod with a pointed tip into a metal specimen with a known force. Two tips; are the readings the same?
- **Experiment 1 (CRD)**: 20 specimens, half tested by tip 1 and half by tip 2, randomly assigned. Use the two-sample t-test. Problem: specimens cut from different bar stock (different heats) are not homogeneous, so specimen-to-specimen variability inflates the error and masks tip differences.
- **Experiment 2 (RCBD / paired)**: divide each of 10 specimens into two halves; randomly assign tip 1 to one half and tip 2 to the other. Each specimen is a *block*.

8.2 Model and the elimination of the block effect

$$y_{ij} = \mu_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, 2;\ j = 1, \ldots, 10,$$

μ_i: true mean hardness of tip i; β_j: effect of specimen j; $\varepsilon_{ij} \overset{\text{iid}}{\sim} (0, \sigma^2)$. Define the paired difference

$$\begin{aligned} d_j = y_{1j} - y_{2j} &= (\mu_1 + \beta_j + \varepsilon_{1j}) - (\mu_2 + \beta_j + \varepsilon_{2j}) \\ &= (\mu_1 - \mu_2) + (\varepsilon_{1j} - \varepsilon_{2j}). \end{aligned}$$

The specimen effect β_j **cancels**. Hence $\mu_d = \mathrm{E}(d_j) = \mu_1 - \mu_2$, $\mathrm{Var}(d_j) = 2\sigma^2$, and $d_1, \ldots, d_n$ is a one-sample problem:

$$t_0 = \frac{\bar{d}}{S_d/\sqrt{n}} \overset{H_0:\mu_d=0}{\sim} t_{n-1}, \qquad \bar{d} = \frac{1}{n}\sum_j d_j, \quad S_d^2 = \frac{1}{n-1}\sum_j (d_j - \bar{d})^2.$$

Remark 2.7: Why pairing gains precision

If instead the 20 specimens were *random* blocks, $\beta_j \sim (0, \sigma_\beta^2)$, then in a CRD $\mathrm{Var}(y) = \sigma_\beta^2 + \sigma^2$ and

$$\mathrm{Var}(\bar{y}_1 - \bar{y}_2)_{\text{CRD}} = \frac{2(\sigma_\beta^2 + \sigma^2)}{n}, \qquad \mathrm{Var}(\bar{d})_{\text{paired}} = \frac{2\sigma^2}{n}.$$

Pairing removes σ_β^2 from the error at the cost of $n - 1$ instead of $2n - 2$ error df. When specimen variability is large, pairing is far more powerful.

Example 2.8: Tip data

Specimen j	1	2	3	4	5	6	7	8	9	10
Tip 1	7	3	3	4	8	3	2	9	5	4
Tip 2	6	3	5	3	8	2	4	9	4	5
d_j	1	0	−2	1	0	1	−2	0	1	−1

$$\begin{aligned} \bar{d} &= \frac{1 + 0 - 2 + 1 + 0 + 1 - 2 + 0 + 1 - 1}{10} = \frac{-1}{10} = -0.1, \\ \sum d_j^2 &= 1 + 0 + 4 + 1 + 0 + 1 + 4 + 0 + 1 + 1 = 13, \\ S_d^2 &= \frac{\sum d_j^2 - n\bar{d}^2}{n-1} = \frac{13 - 10(0.01)}{9} = \frac{12.9}{9} = 1.4333, \quad S_d = 1.1972, \\ t_0 &= \frac{-0.1}{1.1972/\sqrt{10}} = \frac{-0.1}{0.3786} = -0.264. \end{aligned}$$

$|t_0| = 0.264 < t_{0.025,9} = 2.262$ ($P = 0.7976$): fail to reject $H_0 : \mu_d = 0$. No evidence that the two tips give different readings. 95% CI: $-0.1 \pm 2.262(0.3786) = [-0.956, 0.756]$.

9 Nonparametric Alternatives

- **Wilcoxon rank-sum test** (two independent samples): replaces the data by their ranks in the pooled sample.
- **Wilcoxon signed-rank test** (paired data): ranks $|d_j|$ and attaches the signs.
- These are used when normality is doubtful (see also Kruskal–Wallis in Ch. 3).